Graduation project

	الهندسة	كلية:	المنصورة	جامعة:	
	الفصلين الدراسيين	النظام الدراسى:	هندسة الاتصالات والحاسبات		البرنامج:
✓	الساعات المعتمدة		CCE		كود:
هندسة الاتصالات والحاسبات		التخصص الدقيق:	هندسة الاتصالات والحاسبات		التخصص العام:
AI-Based Smart Traffic Management System					عنوان المشروع:
SS): نظام إدارة المرور الذكي المعتمد على الذكاء الاصطناعي					

مشرفي المشروع:Supervisor(s)

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فكرة المشروع:

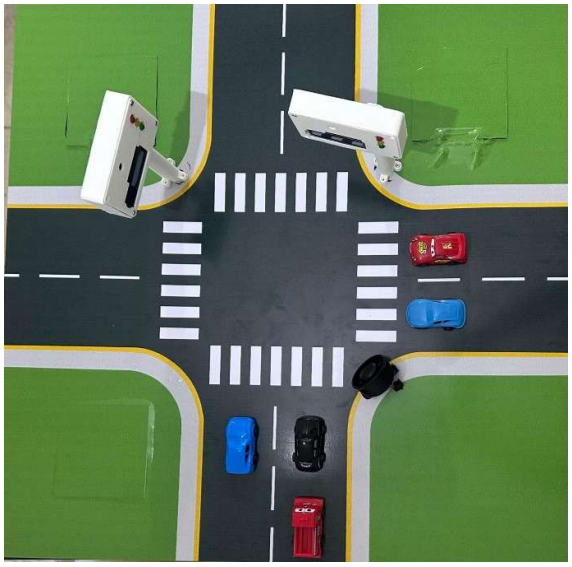
Traditional traffic light systems rely on fixed timers, leading to unnecessary delays, congestion, and severe bottlenecks—especially during emergencies. The **"AI-Based Smart Traffic Management System"** aims to revolutionize intersection control by replacing static timers with adaptive, real-time decision-making.

The system utilizes edge cameras (ESP32-CAM) and an advanced Computer Vision model (YOLOv8) to continuously monitor intersections and dynamically adjust green light durations based on the actual vehicle count in each lane. Furthermore, the project introduces a highly secure **"Emergency Preemption System"** powered by Sensor Fusion. By combining visual ambulance detection with acoustic siren detection (LM393), the system can instantly override normal operations to provide a clear, dedicated path for emergency vehicles, ensuring faster response times while preventing system abuse or false alarms.

أهم النتائج:

- **Dynamic Traffic Optimization:** Successfully transitioned from static timing to adaptive scheduling. The system intelligently allocates longer green light durations to highly congested lanes, significantly reducing average vehicle waiting times and improving overall traffic flow.
- **Foolproof Emergency Response:** Achieved a highly reliable, zero-latency emergency override system. The implementation of a dual-verification condition (Visual confirmation of the ambulance + Acoustic detection of the siren) eliminates false positives and immediately clears the path for life-saving operations.
- **High Fault Tolerance (Robustness):** Designed and implemented a secure **"Fallback Mode"**. In the event of a Wi-Fi disconnection or server failure, the main microcontrollers (ESP32) automatically detect the data loss and revert to a safe, pre-programmed fixed-timer mode, guaranteeing that the intersection never experiences a gridlock.
- **Efficient Distributed Architecture:** Successfully separated heavy AI computations (running on a local Python server) from physical hardware execution (Microcontrollers). This dual-node architecture ensures high-speed data transmission, stability, and scalability for future urban implementation.

صور خاصة بالمشروع:

شكل (2): صور المنتج (المشروع ان وجد)	**(شكل (1): صور من تطبيق المحمول (ان وجد**
	
شكل (4): صورة جماعية لطلاب المشروع	**شكل (3): صفحة الويب الخاصة بالمشروع**

نقاط التميز والتفرد للمشروع :

Dual Sensor Fusion (High Accuracy): Combines advanced computer vision (YOLOv8) with acoustic sensors (LM393) to verify emergencies visually and audibly, completely eliminating false alarms and system abuse.
High Fault Tolerance (Fail-Safe System): Features an automated "Fallback Mode." If the Wi-Fi or AI server fails, the hardware instantly reverts to a traditional fixed-timer cycle, ensuring the intersection never experiences a gridlock.
Distributed Edge Architecture: Strategically separates heavy AI processing (handled by a local Python server) from physical hardware execution (handled by ESP32 microcontrollers), ensuring zero-latency response and maximum stability.
Dynamic Traffic Allocation: Abandons blind, fixed timers in favor of real-time adjustments. The system dynamically allocates green light durations based on live vehicle counts, drastically reducing congestion and waiting times.

Cost-Effective & Scalable: Utilizes highly affordable microcontrollers (ESP32 and ESP32-CAM) as edge nodes. This software-driven approach keeps hardware costs low, making the system highly scalable for real-world urban deployment.